layer of sediment that does not get mixed into contact with seawater and oxidized except possibly during sudden ocean overturns. The last overturn is thought to have been during the most recent ice age, about 20,000 years ago—twenty times longer than the oxidation time in the mixed layer. So the unmixed layer could have twenty times more organic carbon than the mixed layer, and as much as twenty times that on land and in the ocean's surface. If brought to the ocean surface by a new overturn and there oxidized, this is nearly enough to leave everyone gasping for oxygen and dying of CO_2 poisoning; see the end of Chapter 12. Thus such a scenario is conceivable, though highly unlikely.

Chapter 15. Visualizing *Interstellar*'s Wormhole

Christopher Nolan chose several kilometers for the diameter of *Interstellar*'s wormhole. The wormhole's angular diameter as seen from Earth, in radians, is this diameter divided by its distance from Earth, which is about 9 astronomical units or 1.4×10^9 kilometers (the radius of Saturn's orbit). Therefore, the wormhole's angular diameter is about (2 kilometers)/(1.4×10^9 kilometers) = 1.4×10^{-9} radians, which is 0.0003 arc-seconds. Radio telescopes routinely achieve this angular resolution using transworld interferometry. Optical telescopes on the ground using a technique called "adaptive optics," and the Hubble space telescope in space, achieve angular resolutions a hundred times worse than this in 2014. Interferometry between twin Keck telescopes in Hawaii in 2014 can achieve a resolution ten times worse than the wormhole's angular diameter, and it is very plausible that in the era of *Interstellar* optical interferometry between more widely spaced optical telescopes will make possible resolutions better than the wormhole's 0.0003 arc-seconds.